

Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: MMLU | Context: Non-Arab Tourists and Expats in Eight Arab Countries

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU's text-only English MCQ format [Q3, Q56, Q71] matches the deployment's primary interaction modality (English text), which the elicitation rates MODERATE priority. However, the Arabic-learning secondary cohort cannot be evaluated by MMLU at all, and MENAValues research [WEB-11] documents cross-lingual value shifts indicating that English MMLU performance does not predict Arabic-prompt behavior. Format brittleness (e.g., separator-token sensitivity [Q105]) adds modest external-validity concern but is not region-specific.

ID	Evidence
Q3	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
Q56	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
Q71	"Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format." (p.7)
Q105	"If we remove the </s> from the input, accuracy declines by several percentage points." (p.12)
WEB-11	MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU's MCQ accuracy cannot match (https://arxiv.org/html/2510.13154)

Strengths

- English text-only MCQ format aligns with the primary user cohort's interaction modality (Strength 2; D6, D7)
- Zero-shot/few-shot evaluation paradigm [Q3] matches the open-domain generalization required for tourist/expat queries (Strength 4)
- Formal English register is appropriate for educated adult tourist/expat users [Q56]

ID	Evidence
D6	A truck driver was assigned to drive a huge tractor-trailer loaded with logs... the girl was struck by the tractor-trailer and was seriously injured — US tort law scenario
D7	A patient filed a medical malpractice action against a hospital in federal court — US federal court procedure
Q3	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
Q56	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: MMLU is exclusively English text [Q71]; primary deployment interaction is English text — match. Arabic-script and dialect inputs from the secondary cohort are unsupported by MMLU and require separate benchmarking [WEB-1, WEB-2].

IF-2: Regional infrastructure not a concern — tourists/expats use standard text-based devices; deployment YAML notes this is not a connectivity gap context.

IF-3: DialectalArabicMMLU [WEB-1] covers Moroccan, Egyptian, Emirati, Saudi dialects (4 of 8 countries) but not Kuwaiti, Jordanian, Palestinian, or Lebanese dialects; format-level dialect sensitivity is unaddressed in MMLU.

IF-4: Form mismatch is moderate: English MCQ matches primary cohort; Arabic-language input form is entirely unevaluated by MMLU. Cross-lingual value shifts [WEB-11] mean MMLU scores cannot be assumed to transfer to Arabic prompts.

ID	Evidence
Q71	“Existing large-scale NLP models, such as GPT-3, do not incorporate multimodal information, so we design our benchmark to capture a diverse array of tasks in a text-only format.” (p.7)
WEB-1	DialectalArabicMMLU covers five Arabic dialects (Syrian, Egyptian, Emirati, Saudi, Moroccan) with substantial cross-dialect performance variation (https://arxiv.org/abs/2510.27543)
WEB-2	ArabicMMLU is the closest available substitute for the Arabic-language secondary cohort (14,575 MSA questions) (https://aclanthology.org/2024.findings-acl.334/)
WEB-11	MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU’s MCQ accuracy cannot match (https://arxiv.org/html/2510.13154)

Information Gaps

- Tourist/expat device interface preferences (mobile vs. desktop) not searchable at required precision (FV10)
- Whether Arabic-learning users predominantly prompt in MSA vs. dialect

Requires Expert Verification

- Empirical sizing of the Arabic-prompting sub-cohort relative to the English primary cohort

Remediation

Gap 1: English-only format does not evaluate the Arabic-learning sub-cohort.

Recommendation: Add DialectalArabicMMLU [WEB-1] for Moroccan/Egyptian/Emirati/Saudi dialect coverage and ArabicMMLU [WEB-2] for MSA. Note that Kuwaiti, Jordanian, Palestinian, and Lebanese dialects remain unbenchmarked — flag as residual gap.

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